FRC Project Weekly Update

Tilt Baseline, $n = 1$ Diagnostics, and Reproducible Group Release

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Status Since Last Update

Main message

The inherited FRC equilibrium/tilt workflow is now reproduced, operational, quantitatively diagnosed, and packaged as a portable group baseline.

- Legacy $B_{\mathrm{ext}} = 0.13\ \mathrm{T}$ MRR/GS initialization reproduced.
- WarpX + openPMD + ADIOS2/BP5 workflow validated.
- Seeded tilt case extended to 2000 steps:

$$t_{\mathrm{final}} = 2\ \mu\mathrm{s}.$$

- First density-based $n = 1$ displacement diagnostic implemented.
- Portable starter package and offline dependency bundle completed.
- Clean package-only initializer + 200-step smoke test passed.

Validated 2 μs Tilt Baseline

- Reference case:

$$B_{\mathrm{ext}} = 0.13\ \mathrm{T}.$$

- Time step:

$$\Delta t = 10^{-9}\ \mathrm{s}.$$

- Production length:

$$N_{\mathrm{step}} = 2000.$$

- Total physical time:

$$\boxed{t_{\mathrm{final}} = 2\ \mu\mathrm{s}}.$$

- 21 openPMD/BP5 snapshots.
- No recorded WarpX warnings.

Production job

Job	125344674
State	COMPLETED
Elapsed	01:05:50
CPUs	32
Peak memory	$\sim$ 7.25 GB
Output	$\sim$ 3.3 GB

State Evolution During the Baseline

- At $t = 0$:

$$\rho_{\max} \simeq 68.02, \qquad B_{z,\min} \simeq -0.1124\ \mathrm{T}, \qquad B_{z,\max} \simeq 0.2654\ \mathrm{T}.$$

- At $t = 2\ \mu\mathrm{s}$:

$$\rho_{\max} \simeq 66.12, \qquad B_{z,\min} \simeq -0.1105\ \mathrm{T}, \qquad B_{z,\max} \simeq 0.2620\ \mathrm{T}.$$

- Field reversal remains present over the simulated interval.
- Integrated proton charge:

$$\frac{Q(2\ \mu\mathrm{s})}{Q_0} = 0.99631.$$

- Corresponding reduction:

$$\boxed{\Delta Q/Q_0 \simeq 0.37\%}.$$

Interpretation

No evidence of catastrophic particle loss or numerical collapse over the first 2 $\mu\mathrm{s}$.

First Quantitative $n = 1$ Tilt Diagnostic

Complex displacement moment

$$Z_1(t) = \frac{\int z\, \rho_{\mathrm{p}}(\mathbf{x}, t) e^{-i\theta}\, dV}{\int \rho_{\mathrm{p}}(\mathbf{x}, t)\, dV}, \qquad A_1(t) = |Z_1(t)|.$$

- Full-density and 5%-core definitions give comparable amplitudes.
- Reference radius:

$$R_0 = 0.1706747 \text{ m}.$$

- Initial amplitude:

$$A_1(0) \simeq 0.0033 \text{ mm}.$$

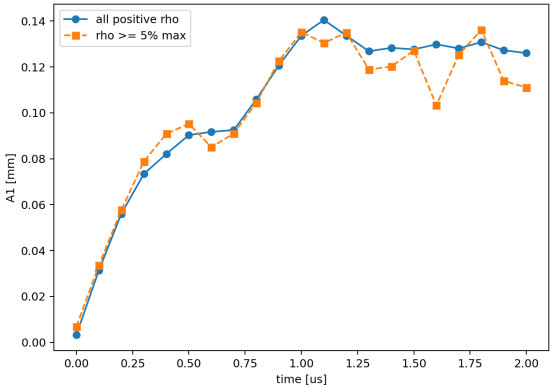
- Maximum observed amplitude:

$$\boxed{A_1^{\max} \simeq 0.141 \text{ mm}}$$

near

$$t \simeq 1.1\ \mu\text{s}.$$

Tilt Evolution: Current Interpretation



$n = 1$ displacement amplitude

- Initial response rises to

$$A_1 \sim 0.14 \text{ mm}.$$

- Later amplitude remains approximately

$$0.126\text{–}0.131 \text{ mm}.$$

- Normalized late-time displacement:

$$\frac{A_1}{R_0} \sim 7.5 \times 10^{-4}.$$

- No clear sustained exponential interval is visible yet.

Current conclusion

The first 2 μs characterize the seeded response, but do not yet justify extraction of a physical tilt

Why γ_{tilt} Is Not Reported Yet

Expected linear regime

$$A_1(t) \propto e^{\gamma t} \qquad \Longrightarrow \qquad \ln A_1(t) = \gamma t + C.$$

The present evolution is better summarized as

$$\boxed{\text{seeded response} \rightarrow \text{rise to } A_1 \sim 0.14\ \mathrm{mm} \rightarrow \text{approximate plateau}}.$$

- A full-interval exponential fit would not yet be physically defensible.
- The early increase may include seed response and plasma relaxation.
- A second geometric diagnostic is required before interpreting the signal as rigid-body tilt.

$$\boxed{\gamma_{\mathrm{tilt}} \text{ not yet extracted}}$$

Next Diagnostic: FRC Centerline Tilt

Transverse center of charge

$$x_c(z) = \frac{\int x\rho\, dx\, dy}{\int \rho\, dx\, dy}, \qquad y_c(z) = \frac{\int y\rho\, dx\, dy}{\int \rho\, dx\, dy}.$$

For an approximately rigid tilt,

$$x_c(z) \simeq a_x z + b_x, \qquad y_c(z) \simeq a_y z + b_y.$$

A geometric tilt angle can then be estimated as

$$\boxed{\theta_{\mathrm{tilt}} \simeq \tan^{-1}\left(\sqrt{a_x^2 + a_y^2}\right)}.$$

- Provides a direct geometric interpretation.
- Can be compared against the Fourier-like A_1 diagnostic.
- If the plateau persists, extend the baseline toward 5 μs.

Portable FRC Simulation Package

Main release

`FRC_Simulation_Starter_Pack_v1.0_full.zip`

- Original inherited Wei Qi initializer/input preserved under `legacy/`.
- Reproduced 0.13 T reference case.
- Portable Slurm scripts for:
  - initialization,
  - 200-step smoke test,
  - 2000-step tilt baseline,
  - $n = 1$ post-processing.
- ADIOS2/WarpX build scripts and compatibility patch.
- Reference $n = 1$ diagnostic output.
- Environment, software, and job provenance.
- Setup, usage, known-issues, and team-interface documentation.
- `MANIFEST.txt` and SHA256 checksums.

Offline Reconstruction Bundle

Offline source package

`FRC_Simulation_Offline_Sources_v1.0.tar.gz`

Contains the exact source snapshots used for:

- WarpX,
- openPMD-api,
- ADIOS2,
- AMReX,
- PICSAR,
- nlohmann_json,
- toml11.

Purpose

The validated software environment can be reconstructed without relying on external downloads from HPC compute nodes.

Clean Release Certification

Independent validation path

portable package $\rightarrow$ 0.13 T initializer $\rightarrow$ `init.bp` $\rightarrow$ 200-step WarpX/ADIOS2 smoke

Initializer

Job	125580121
State	COMPLETED
Exit code	0
Elapsed	00:01:37

WarpX smoke

Job	125580123
State	COMPLETED
Exit code	0
Elapsed	00:07:26

```
=== SMOKE TEST PASS ===
```

Release Integrity and Attribution

- Active scripts no longer depend on:
  - personal username paths,
  - historical `gs_tilt/runs/` paths.
- Exact historical scripts are retained separately under `provenance/reference_scripts/`.
- All internal SHA256 checks passed.
- ZIP and TAR archive integrity checks passed.
- The reference simulation remains explicitly attributed to **Wei Qi ()**.
- Present contribution:

$$\text{reproduction} + \text{environment reconstruction} + \text{validation} + \text{diagnostics} + \text{packaging}.$$

Group Handoff

Files prepared for the group

- `FRC_Simulation_Starter_Pack_v1.0_full.zip`
- `FRC_Simulation_Offline_Sources_v1.0.tar.gz`
- `FRC_Simulation_v1.0_SHA256.txt`

- Wei Qi was contacted first because the reference baseline originates from his previous work.
- He approved sharing the package with the wider group.
- The group now has a common reproducible baseline rather than depending on one working directory.

inherited workflow $\rightarrow$ shared group baseline

Current Status

Task	Status
Legacy 0.13 T MRR/GS reproduction	DONE
Inherited $n = 1$ seed reproduction	DONE
WarpX/openPMD/ADIOS2 reconstruction	DONE
200-step smoke validation	DONE
2000-step / 2 μs tilt baseline	DONE
First density-based $n = 1$ diagnostic	DONE
Portable group package	DONE
Clean release certification	DONE
Offline source bundle	DONE
Wei Qi approval for group sharing	DONE
Centerline tilt diagnostic	NEXT
Longer tilt evolution	NEXT
Geometry/stability parameter study	NEXT

Next Research Steps

1. Implement the centerline-based geometric tilt diagnostic.
2. Compare
$$\theta_{\mathrm{tilt}}(t) \qquad \text{and} \qquad A_1(t).$$
3. If the plateau persists, continue the baseline from the 2 μs checkpoint toward approximately
$$t \simeq 5\ \mu\mathrm{s}.$$
4. Extract γ_{tilt} only after identifying a defensible linear interval in $\ln A_1(t)$.
5. Begin connecting tilt behavior to device-design variables:
$$R_s, \quad L_s, \quad E, \quad R_w/R_s, \quad B_{\mathrm{ext}}, \quad S^*.$$

Research Direction

Transition

The reproduction and infrastructure phase is now substantially closed.

$$\boxed{\text{inherited files} \rightarrow \text{reproduced equilibrium} \rightarrow \text{validated dynamics} \rightarrow \text{tilt diagnostics} \rightarrow \text{shared baseline}}$$

The next scientific focus is

$$\boxed{\text{FRC equilibrium/tilt stability} \quad + \quad \text{realistic device geometry/design}}.$$

How do FRC and device dimensions affect tilt stability?

Request for Feedback

- Is the centerline tilt angle the right next cross-check for $A_1(t)$?
- If the plateau persists, should the baseline be extended directly to 5 μs?
- Which device parameter should be scanned first after the reference tilt behavior is established:

$$E, \qquad R_w/R_s, \qquad B_{\mathrm{ext}}, \qquad S^*?$$